

ECO - LOOP BUILDING AGENTS

Problem Statement ID	1
Problem Statement Title	ECO - LOOP BUILDING AGENTS
Theme	
PS Category	Software
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Eco-Loop: LLM-Driven Closed-Loop HVAC Control

Detailed Explanation

A supervisory AI layer atop a building's HVAC system, replacing fixed-schedule setpoints with dynamic decisions. Runs real EnergyPlus simulation (DOE Reference Medium Office, 18 zones, Chicago climate) connected to a local LLM (Llama 3.2 via Ollama) through a validated tool-calling interface. Every simulated hour, the LLM reads live zone temperature, PMV comfort, and outdoor conditions, then proposes setpoints validated against ASHRAE 55 constraints before being written back via the EnergyPlus Python Actuator API.

Problem Addressed

Traditional BMS use static setpoint schedules that cannot adapt to real-time occupancy, comfort feedback, or grid conditions. Eco-Loop closes this gap with a continuous reasoning layer that adjusts setpoints based on live conditions rather than a fixed calendar.

Innovation & Uniqueness

Unlike black-box RL controllers, Eco-Loop's LLM reasoning is interpretable and safety-bounded. An explicit validation layer rejects unsafe proposals and falls back to the last known-good setpoint if LLM output is malformed — stress-tested during development, including observed real recovery from malformed output.

Implementation Stack & Closed-Loop Methodology

Technology Stack

- **Python 3.12**
- **EnergyPlus 26.1** (pyenergyplus) — industry-standard building energy simulation engine
- **DOE Commercial Reference Building** (Medium Office, 18 zones) + Chicago TMY3 weather
- **Ollama + Llama 3.2** — open-source, locally hosted LLM, no cloud dependency
- **ASHRAE 55 / Fanger PMV** — occupant thermal comfort quantification



The four-stage loop executes once per simulated hour, with a zero-order hold between decision calls to bound latency and cost.

Validated End-to-End on Standard Hardware

Feasibility Analysis

- Full pipeline built and validated on a standard laptop — no specialized hardware or cloud dependency
- Tested over a full summer peak week (07/14–07/20, Chicago) comparing baseline fixed-schedule vs. live LLM control
- Results: **–5.8% total electricity** and comfort violations reduced from **40.0% → 23.1%** of timesteps ($|PMV| > 0.5$) — energy and comfort improved simultaneously

Risks & Mitigations

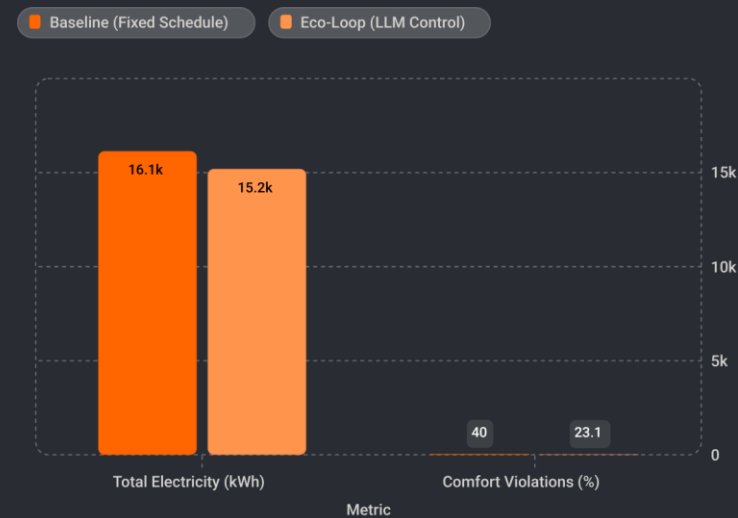
- **Malformed LLM output:** Validation + fallback layer rejects bad proposals, holds last known-good setpoint
- **Inference latency:** Calls throttled to once per simulated hour (zero-order hold), bounding wall-clock delay
- **Single shared setpoint pair:** Per-zone control is a natural extension — same tool-calling interface generalizes to per-zone actuators

Deliverables & Results Dashboard

Submitted Artifacts

- **Source Code:** Python modules — EnergyPlus controller, tool-calling validation layer, LLM decision interface — full repository on GitHub
- **Live Run Snapshots:** Terminal output showing live sensor readings, LLM decisions, and validation results across a simulated week, including graceful recovery from a malformed LLM response
- **Results Dashboard:** Comparative bar chart of baseline vs. AI-controlled performance over the same weather week

Energy & Comfort Results



Energy reduced by **-5.8%** (16,142 → 15,199 kWh) and comfort violations reduced by **16.9 points** (40.0% → 23.1%) over the same summer peak week.



Placeholder: Insert terminal output screenshots and live-run validation log snapshots here.

Research & References

01

EnergyPlus Simulation Platform

U.S. Department of Energy, EnergyPlus Simulation Software — <https://energyplus.net>

03

DOE Commercial Reference Building Models

Deru, M. et al. (2010), U.S. DOE Commercial Reference Building Models of the National Building Stock, NREL/TP-5500-46861

05

Local LLM Runtime & Model

Ollama, Local Large Language Model Runtime — <https://ollama.com>;
Meta AI, Llama 3.2 Model Card — <https://ai.meta.com/llama>

02

EnergyPlus Python API

NREL, EnergyPlus Python API Documentation — <https://energyplus.readthedocs.io>

04

Thermal Comfort Standards

ASHRAE Standard 55 — Thermal Environmental Conditions for Human Occupancy; Fanger, P.O. (1970), *Thermal Comfort: Analysis and Applications in Environmental Engineering*

06

Model Context Protocol

Model Context Protocol (MCP) Specification, Anthropic — <https://modelcontextprotocol.io>